MA1522 Linear Algebra for Computing
L2 Lesson 2



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Question 1

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Solve the following linear system: $\begin{cases} x - 4y + 2z = -2 \\ x + 2y - 2z = -3 \\ x - y = 4 \end{cases}$

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Solution.

$$\left(\begin{array}{ccc|c} 1 & -4 & 2 & -2 \\ 1 & 2 & -2 & -3 \\ 1 & -1 & 0 & 4 \end{array}\right) \xrightarrow[R_3-R_1]{R_2-R_1} \left(\begin{array}{ccc|c} 1 & -4 & 2 & -2 \\ 0 & 6 & -4 & -1 \\ 0 & 3 & -2 & 6 \end{array}\right)$$

$$\xrightarrow{R_3-\frac{1}{2}R_2} \left(\begin{array}{ccc|c} 1 & -4 & 2 & -2 \\ 0 & 6 & -4 & -1 \\ 0 & 0 & 0 & \frac{13}{2} \end{array}\right)$$

The rightmost column of a row-echelon form is pivot. Thus, the system is inconsistent, i.e. has no solution. □

Consistency of a Linear System

Let $\boldsymbol{A}$ be the augmented matrix of a linear system. Apply Gaussian elimination to get a row-echelon form $\boldsymbol{R}$ of $\boldsymbol{A}$.

1. If the rightmost column of R is **pivot**, then the system is **inconsistent**.
2. If the rightmost column of R is **non-pivot**, then the system is **consistent**.
   a. If all other columns are pivot, then system has **exactly one** solution.
   b. If some other column is non-pivot, then the system has **infinitely many** solutions.

Number of non-pivot columns (excluding the rightmost one) = number of arbitrary parameters.

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Question 2

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Determine if each of the following statements is true or false:

a. There is an inconsistent linear system with more variables than equations.

b. There is a linear system with more variables than equations that has exactly one solution.

c. There is a linear system with more equations than variables that has exactly one solution.

d. There is a linear system with more equations than variables that has infinitely many solutions.

e. There is a linear system with three unknowns that has infinitely many solutions with three arbitrary parameters.

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e. There is a linear system with three unknowns that has infinitely many solutions with three arbitrary parameters. . . . . **Ans: True**

Question 2: Solution

Let $\boldsymbol{R}$ be a row-echelon form of the augmented matrix of a linear system. Then

$$\begin{aligned}\text{number of variables} &= \text{number of columns of } \boldsymbol{R} - 1;\\ \text{number of equations} &= \text{number of rows of } \boldsymbol{R}.\end{aligned}$$

Recall that the linear system is consistent if and only if the rightmost column of $\boldsymbol{R}$ is non-pivot, in which case

$$\begin{aligned}&\text{Number of non-pivot columns of } \boldsymbol{R} \text{ (excluding the rightmost one)}\\ &= \text{number of arbitrary parameters in the solutions.}\end{aligned}$$

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Question 2(a): Solution

a. There is an inconsistent linear system with more variables than equations.

For a linear system to be inconsistent, we need the rightmost column of R to be pivot, and for a linear system to have more variables than equations, we need

$$\begin{aligned}\text{number of columns of } R - 1 &= \text{number of variables} \\ &> \text{number of equations} = \text{number of rows of } R.\end{aligned}$$

Examples of such R:

$$\left(\begin{array}{cc|c} 0 & 0 & 1 \end{array}\right); \quad \left(\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right).$$

Answer: True

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For a linear system to be inconsistent, we need the rightmost column of $\boldsymbol{R}$ to be pivot, and for a linear system to have more variables than equations, we need

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Answer: True

Question 2(b,c): Solution

b. There is a linear system with more variables than equations that has exactly one solution.

c. There is a linear system with more equations than variables that has exactly one solution.

For a system to have exactly one solution, we need the rightmost column of R to be the only non-pivot column. Therefore,

$$\begin{aligned}\text{number of variables} &= \text{number of columns of } R - 1\\ &= \text{number of pivot columns of } R\\ &= \text{number of leading entries of } R\\ &= \text{number of nonzero rows of } R\\ &\leq \text{number of rows of } R\\ &= \text{number of equations.}\end{aligned}$$

Answer: (b) False. (c) True. Example of R: $\left(\begin{array}{c|c} 1 & 1 \\ 0 & 0 \end{array}\right)$.

Question 2(b,c): Solution

(b) There is a linear system with more variables than equations that has exactly one solution.

(c) There is a linear system with more equations than variables that has exactly one solution.

For a system to have exactly one solution, we need the rightmost column of $\boldsymbol{R}$ to be the only non-pivot column. Therefore,

$$\begin{aligned}
\text{number of variables} &= \text{number of columns of } \boldsymbol{R} - 1 \\
&= \text{number of pivot columns of } \boldsymbol{R} \\
&= \text{number of leading entries of } \boldsymbol{R} \\
&= \text{number of nonzero rows of } \boldsymbol{R} \\
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&= \text{number of equations.}
\end{aligned}$$

Answer: (b) False. (c) True. Example of $\boldsymbol{R}$: $\left(\begin{array}{c|c} 1 & 1 \\ 0 & 0 \end{array}\right)$.

Question 2(b,c): Solution

b. There is a linear system with more variables than equations that has exactly one solution.

c. There is a linear system with more equations than variables that has exactly one solution.

For a system to have exactly one solution, we need the rightmost column of $\boldsymbol{R}$ to be the only non-pivot column. Therefore,

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Answer: (b) False. **(c) True.** Example of $\boldsymbol{R}$: $\left(\begin{array}{c|c} 1 & 1 \\ 0 & 0 \end{array}\right)$.

Question 2(d): Solution

d. There is a linear system with more equations than variables that has infinitely many solutions.

For a system to have infinitely many solutions, R would have at least two non-pivot columns which include the rightmost one. An example of such R is $\left(\begin{array}{cc|c} 1 & 0 & 1 \end{array}\right)$. Adding two zero rows to this matrix will give the desired linear system that has more equations that variables.

Answer: True. Example of R: $\left(\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right)$.

Question 2(d): Solution

(d) There is a linear system with more equations than variables that has infinitely many solutions.

For a system to have infinitely many solutions, $\boldsymbol{R}$ would have at least two non-pivot columns which include the rightmost one. An example of such $\boldsymbol{R}$ is $\left(\begin{array}{cc|c} 1 & 0 & 1 \end{array}\right)$. Adding two zero rows to this matrix will give the desired linear system that has more equations that variables.

Answer: True. Example of $\boldsymbol{R}$: $\left(\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right)$.

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Answer: True. Example of $\boldsymbol{R}$: $\left(\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right)$.

Question 2(e): Solution

(e) There is a linear system with three variables that has infinitely many solutions with three arbitrary parameters.

If a system has infinitely many solutions with three arbitrary parameters, then it would have at least four non-pivot columns which include the rightmost one. Since R in this case has only four columns, all columns are non-pivot, and hence zero.

Answer: True. Example of R : $\left(\begin{array}{ccc|c} 0 & 0 & 0 & 0 \end{array}\right)$ corresponding to the linear system $0x + 0y + 0z = 0$.

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Question 3

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Find the values of a and b such that the following linear system in variables x, y, z

$$\begin{cases} ax + bz = 2 \\ ax + ay + 4z = 4 \\ ay + 2z = b \end{cases}$$

has

a. exactly one solution;
b. infinitely many solutions with one arbitrary parameter;
c. infinitely many solutions with two arbitrary parameters;
d. no solution.

Question 3: Solution

$$\begin{cases} ax + bz = 2 \\ ax + ay + 4z = 4 \\ ay + 2z = b \end{cases}. \qquad \text{Augmented matrix } \boldsymbol{A} = \left(\begin{array}{ccc|c} a & 0 & b & 2 \\ a & a & 4 & 4 \\ 0 & a & 2 & b \end{array}\right).$$

$$\boldsymbol{A} \xrightarrow{R_2 - R_1} \left(\begin{array}{ccc|c} a & 0 & b & 2 \\ 0 & a & 4-b & 2 \\ 0 & a & 2 & b \end{array}\right) \xrightarrow{R_3 - R_2} \left(\begin{array}{ccc|c} a & 0 & b & 2 \\ 0 & a & 4-b & 2 \\ 0 & 0 & b-2 & b-2 \end{array}\right) = \boldsymbol{B}$$

To continue we need to know if B is in row-echelon form.

Question 3: Solution

$$\begin{cases} ax + bz = 2 \\ ax + ay + 4z = 4 \\ ay + 2z = b \end{cases}. \qquad \text{Augmented matrix } \boldsymbol{A} = \left(\begin{array}{ccc|c} a & 0 & b & 2 \\ a & a & 4 & 4 \\ 0 & a & 2 & b \end{array}\right).$$

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To continue we need to know if B is in row-echelon form.

Question 3: Solution: Case 1: $a \neq 0$

Then $\boldsymbol{B} = \left(\begin{array}{ccc|c} a & 0 & b & 2 \\ 0 & a & 4-b & 2 \\ 0 & 0 & b-2 & b-2 \end{array}\right)$ is in row-echelon form.

The first two columns of B are pivot columns,
the third column of B is pivot if and only if $b - 2 \neq 0$,
while the last column is never pivot.

Thus, the system is always consistent, and has

$$\begin{cases} \text{exactly one solution,} & \text{if } (a \neq 0 \text{ and}) \ b \neq 2; \\ \text{infinitely many solutions} \\ \text{with one arbitrary parameter}, & \text{if } (a \neq 0 \text{ and}) \ b = 2. \end{cases}$$

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The first two columns of $\boldsymbol{B}$ are pivot columns,
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Question 3: Solution: Case 2: $a = 0$

$$\boldsymbol{B} = \left(\begin{array}{ccc|c} 0 & 0 & b & 2 \\ 0 & 0 & 4-b & 2 \\ 0 & 0 & b-2 & b-2 \end{array}\right) \xrightarrow[\frac{1}{4}R_1]{R_1+R_2} \left(\begin{array}{ccc|c} 0 & 0 & 1 & 1 \\ 0 & 0 & 4-b & 2 \\ 0 & 0 & b-2 & b-2 \end{array}\right)$$

$$\xrightarrow[R_3+(2-b)R_1]{R_2+(b-4)R_1} \left(\begin{array}{ccc|c} 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & b-2 \\ 0 & 0 & 0 & 0 \end{array}\right) = \boldsymbol{C}$$

The first two columns of C are non-pivot, the third column of C is pivot, while the last column is pivot if and only if $b - 2 \neq 0$.

Thus, the system is consistent if and only if $b - 2 = 0$ when $a = 0$, in which case it has infinitely many solutions with two arbitrary parameters.

Question 3: Solution: Case 2: $a = 0$

$$\boldsymbol{B} = \left(\begin{array}{ccc|c} 0 & 0 & b & 2 \\ 0 & 0 & 4-b & 2 \\ 0 & 0 & b-2 & b-2 \end{array}\right) \xrightarrow[\frac{1}{4}R_1]{R_1+R_2} \left(\begin{array}{ccc|c} 0 & 0 & 1 & 1 \\ 0 & 0 & 4-b & 2 \\ 0 & 0 & b-2 & b-2 \end{array}\right)$$

$$\xrightarrow[R_3+(2-b)R_1]{R_2+(b-4)R_1} \left(\begin{array}{ccc|c} 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & b-2 \\ 0 & 0 & 0 & 0 \end{array}\right) = \boldsymbol{C}$$

The first two columns of $\boldsymbol{C}$ are non-pivot, the third column of $\boldsymbol{C}$ is pivot, while the last column is pivot if and only if $b - 2 \neq 0$.

Thus, the system is consistent if and only if $b - 2 = 0$ when $a = 0$, in which case it has infinitely many solutions with two arbitrary parameters.

Question 3: Solution: Case 2: $a = 0$

$$\boldsymbol{B} = \left(\begin{array}{ccc|c} 0 & 0 & b & 2 \\ 0 & 0 & 4-b & 2 \\ 0 & 0 & b-2 & b-2 \end{array}\right) \xrightarrow[\frac{1}{4}R_1]{R_1+R_2} \left(\begin{array}{ccc|c} 0 & 0 & 1 & 1 \\ 0 & 0 & 4-b & 2 \\ 0 & 0 & b-2 & b-2 \end{array}\right)$$

$$\xrightarrow[R_3+(2-b)R_1]{R_2+(b-4)R_1} \left(\begin{array}{ccc|c} 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & b-2 \\ 0 & 0 & 0 & 0 \end{array}\right) = \boldsymbol{C}$$

The first two columns of $\boldsymbol{C}$ are non-pivot, the third column of $\boldsymbol{C}$ is pivot, while the last column is pivot if and only if $b - 2 \neq 0$.

Thus, the system is consistent if and only if $b - 2 = 0$ when $a = 0$, in which case it has infinitely many solutions with two arbitrary parameters.

Question 3: Final Solution

The linear system $\begin{cases} ax + bz = 2 \\ ax + ay + 4z = 4 \\ ay + 2z = b \end{cases}$ has

a. exactly one solution **when** $a \neq 0$ **and** $b \neq 2$;

b. infinitely many solutions with one arbitrary parameter **when** $a \neq 0$ **and** $b = 2$;

c. infinitely many solutions with two arbitrary parameters **when** $a = 0$ **and** $b = 2$;

d. no solution **when** $a = 0$ **and** $b \neq 2$.

Question 4

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Solve the following quadratic system: $\begin{cases} x^2 - y^2 + 2z^2 = 6 \\ 2x^2 + 2y^2 - 5z^2 = 3 \\ 2x^2 + 5y^2 + z^2 = 9 \end{cases}$.

Question 4

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Solve the following quadratic system: $\begin{cases} x^2 - y^2 + 2z^2 = 6 \\ 2x^2 + 2y^2 - 5z^2 = 3 \\ 2x^2 + 5y^2 + z^2 = 9 \end{cases}$.

Solution.

Let $a = x^2$, $b = y^2$, $c = z^2$.

Then the quadratic system becomes a linear one: $\begin{cases} a - b + 2c = 6 \\ 2a + 2b - 5c = 3 \\ 2a + 5b + c = 9 \end{cases}$.

Question 4

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Solve the following quadratic system: $\begin{cases} x^2 - y^2 + 2z^2 = 6 \\ 2x^2 + 2y^2 - 5z^2 = 3 \\ 2x^2 + 5y^2 + z^2 = 9 \end{cases}$.

Solution.

Let $a = x^2$, $b = y^2$, $c = z^2$.

Then the quadratic system becomes a linear one: $\begin{cases} a - b + 2c = 6 \\ 2a + 2b - 5c = 3 \\ 2a + 5b + c = 9 \end{cases}$.

$$\left(\begin{array}{ccc|c} 1 & -1 & 2 & 6 \\ 2 & 2 & -5 & 3 \\ 2 & 5 & 1 & 9 \end{array}\right) \xrightarrow{\text{Gauss-Jordan}} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{array}\right)$$

Question 4

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Solve the following quadratic system: $\begin{cases} x^2 - y^2 + 2z^2 = 6 \\ 2x^2 + 2y^2 - 5z^2 = 3 \\ 2x^2 + 5y^2 + z^2 = 9 \end{cases}$.

Solution.

Let $a = x^2$, $b = y^2$, $c = z^2$.

Then the quadratic system becomes a linear one: $\begin{cases} a - b + 2c = 6 \\ 2a + 2b - 5c = 3 \\ 2a + 5b + c = 9 \end{cases}$.

$$\left(\begin{array}{ccc|c} 1 & -1 & 2 & 6 \\ 2 & 2 & -5 & 3 \\ 2 & 5 & 1 & 9 \end{array}\right) \xrightarrow{\text{Gauss-Jordan}} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{array}\right)$$

Thus $a = 4$, $b = 0$, $c = 1$;

and so $x = \pm 2$, $y = 0$, $z = \pm 1$. □

Next Lesson ...

We will explore **Sections 2.1–2.3**.
Please watch/review the video clips for these sections before then.